Lecture 4

Log is defined only on $(0,+\infty)$

determine whether the following limits exist

$\lim_{(x,y)\to(0,0)} e^{-x-y} = 1$ ✓

$\lim_{x\to0}(2x)=0$, $\lim_{x\to0}(-x)=0$, $\lim_{y\to0}(-y)=0$ (inner function)

$\lim_{(x,y)\to(0,0)} x^2+y^2=0$ ✓

$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2}=0$ ✓

$\left|\frac{x^2y}{x^2+y^2}-0\right| = \left|\frac{x^2y}{x^2+y^2}\right| < \varepsilon$

$\left|\frac{x^2y}{x^2+y^2}\right| \le \left|\frac{(x^2+y^2)y}{x^2+y^2}\right| = |y| \le r$

in other term $|y|<\delta$

because of this $\|(x,y)-(0,0)\| = \sqrt{x^2+y^2} = r$

if we choose $\delta=\varepsilon$ so that

$\|(x,y)-(0,0)\| = \sqrt{x^2+y^2} < \delta = \varepsilon \iff x^2+y^2<\delta^2=\varepsilon^2$

this implies that $|y|<\varepsilon$ ☑ Qed

$f(x,y)=x^2+y^2$

$\lim_{(x,y)\to(0,0)} f(x,y)=0$

Let $\varepsilon>0$ then $|f(x)-0| = |f(x)| = |x^2+y^2|$

Notice that $\|(x,y)-(0,0)\| = \sqrt{x^2+y^2}$

chose $\delta<\sqrt{\varepsilon}$

then $\sqrt{x^2+y^2}<\delta$ implies $x^2+y^2<\varepsilon$

$\|(x,y)-(0,0)\|<\delta \Rightarrow |f(x)-0|<\varepsilon$

Definition Let $f: U\subset\mathbb{R}^n\to\mathbb{R}$. The graph of f is the subset of $\mathbb{R}^{n+1}$ given by $\{x_1,\dots,x_n,f(x_1,\dots,x_n)\}$.

- $n=1$: graph is a curve in $\mathbb{R}^2$.
- $n=2$: graph is a surface in $\mathbb{R}^3$.
- $n=3$: graph is a hypersurface in $\mathbb{R}^4$.

Definition For $f(x,y)$, a **cross-section** is obtained by holding one variable fixed and letting the other vary.

Graphically, it is the intersection of the surface with a plane $x=c$ or $y=c$.

Definition A **level set** is a subset of the domain where f is constant: $f(x_1,\dots,x_n)=c$

A few level sets of the function $f(x,y)=x^2+y^2$ are graphed below.

Definition A function $f:\mathbb{R}^n\to\mathbb{R}$ is **linear** if it can be written as $f(x_1,x_2,\dots,x_n)=c+m_1x_1+m_2x_2+\dots+m_nx_n$, where $c,m_1,\dots,m_n$ are constants.

A convenient formula If a plane has slope m in the x-direction, has slope n in the y-direction, and passes through the point (x_0,y_0,z_0), then its equation is $z=z_0+m(x-x_0)+n(y-y_0)$.

Definition Let $f:\mathbb{R}^n\to\mathbb{R}$ and $\mathbf{a}=(a_1,\dots,a_n)$. We say $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})=L$ if for every $\varepsilon>0$, there exists $\delta>0$ such that $0<\|\mathbf{x}-\mathbf{a}\|<\delta \implies |f(\mathbf{x})-L|<\varepsilon$.

Lecture 5

definition of continuation

Definition
f is continuous at $\mathbf{a}$ if $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$.
A function is continuous on a region U in $\mathbb{R}^n$ if it is continuous at each point in U. *There is no break in the graph of f!*

Using the limit laws, one can show that for functions of several variables, the following are continuous
- sums of continuous functions
- products of continuous functions
- compositions of continuous functions
- quotient of two continuous functions wherever the denominator function is nonzero.

examples

Is the function continuous in the given region?
1. $\frac{1}{x^2+y^2}$ on $-1\le x,y\le1$
2. $\frac{1}{x^2+y^2}$ on $1\le x,y\le2$
3. $\frac{y}{x^2+2}$ on $x^2+y^2\le1$

① $\frac{1}{x^2+y^2}$ on $-1\le x,y\le1$

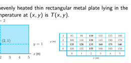
at $x=y=0$ $x^2+y^2=0$

So not continuous since $\lim_{(x,y)\to(0,0)}\frac{1}{x^2+y^2}=DNE$

② $x^2+y^3\le1$

$\sqrt{x^2+y^2}\le1$ dist. of (x,y) to $(0,0)$ being ≤1

Imagine an unevenly heated thin rectangular metal plate lying in the xy-plane. Temperature at (x,y) is $T(x,y)$.



y (m)						
3	85	90	110	135	155	180
2	100	110	120	145	190	170
1	125	128	135	160	175	160
0	120	135	155	160	160	150
	0	1	2	3	4	5

How does T vary near the point $(2,1)$?

along $x=2$

avarage rate of change $\frac{\Delta T}{\Delta y} = \frac{155-135}{0-1} = -20\frac{F}{m}$

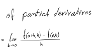
along $y=1$

avarage rate of change $\frac{\Delta T}{\Delta x} = \frac{135-128}{2-1} = 7\frac{F}{m}$

definition of partial derivatives

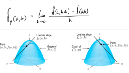
$$f_x(a,b)=\lim_{h\to0}\frac{f(a+h,b)-f(a,b)}{h}$$

$$f_y(a,b)=\lim_{h\to0}\frac{f(a,b+h)-f(a,b)}{h}$$



Partial derivatives are the slope of tangent lines of the cross section.

Sign of a partial derivative



Sign of $f_x(0,5)$? $f_x(0,5)>0$

Sign of $f_y(0,5)$? $f_y(0,5)=0$

note that $f_x=\frac{\partial f}{\partial x}$ & $f_y=\frac{\partial f}{\partial y}$

Plane tangent

at (a,b), if f is differentiable

$$z=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$$

find the tangent line at $(3,4)$ for the plane $f(x,y)=x^2+y^2$

$f(a,b): f(3,4)=25$

$f_x(x,y)=2x: f_x(3,4)=6$

$f_y(x,y)=2y: f_y(3,4)=8$

$L(x,y)=25+6(x-3)+8(y-4)$

Error function:

$E(x,y)=f(x,y)-L(x,y)$

the error is given by $E(x+h,y+k)$

examine distance between $(x+h,y+k)$ and (x,y)

Compare $|E(x+h,y+k)|$ with h^2+k^2

$$\frac{|E(x+h,y+k)|}{h^2+k^2}\approx c$$

note: if f has continuous second derivatives, then

$$f(x,y)-L(x,y)=O(\|(x,y)-(a,b)\|^2)$$

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Using the limit laws, one can show that for functions of several variables, the following are continuous
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- quotient of two continuous functions wherever the denominator function is nonzero.

example:

compute pa of

$$f(x_1\dots x_n)=\sum_{i=1}^n x_i^2$$

$$\frac{\partial f}{\partial x_i}=2x_i$$

Tangent hyperplane approximation/ First order approximation
For $\mathbf{x}=(x_1,\dots,x_n)$ near $\mathbf{a}=(a_1,\dots,a_n)$:
$$f(\mathbf{x})\approx L(\mathbf{x})=f(\mathbf{a})+\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\mathbf{a})(x_i-a_i)$$
Error behavior: If f has continuous second derivatives, then $f(\mathbf{x})-L(\mathbf{x})=O(\|\mathbf{x}-\mathbf{a}\|^2)$.

unit vector $= u = u_1 i + u_2 j$ where $i=(1,0)$ and $j=(0,1)$

note when u is Not a unit vector use $\frac{\vec{u}}{\|\vec{u}\|}$

the average rate of change at P is

$$\frac{\text{change in } f}{\text{Distance of } P,Q} = \frac{f(a+hu_1,b+hu_2)-f(a,b)}{h}$$

now use limit to find the instantaneous roc

$$f_u(a,b)=\lim_{h\to0}\frac{f(a+hu_1,b+hu_2)-f(a,b)}{h}$$

Partial derivatives of f at $(x_1,\dots,x_n)$
For all points at which the limit exist, we define the partial derivatives at the point $(x_1,\dots,x_n)$ by
$$\frac{\partial f}{\partial x_i}(x_1,\dots,x_n)=\lim_{h\to0}\frac{f(x_1,\dots,x_i+h,\dots,x_n)-f(x_1,\dots,x_i,\dots,x_n)}{h}$$
- Compute by treating all other variables as constants.
- Example: $f(x,y,z)=x^2yz$, $\frac{\partial f}{\partial x}=2xyz$, $\frac{\partial f}{\partial y}=x^2z$, $\frac{\partial f}{\partial z}=x^2y$

Gradient
Let $f:U\subset\mathbb{R}^n\to\mathbb{R}$ be differentiable at $\mathbf{a}=(a_1,\dots,a_n)$. The **gradient** of f at $\mathbf{a}$ is
$$\nabla f(\mathbf{a})=\left(\frac{\partial f}{\partial x_1}(\mathbf{a}),\frac{\partial f}{\partial x_2}(\mathbf{a}),\dots,\frac{\partial f}{\partial x_n}(\mathbf{a})\right).$$

$L(\vec{x})=f(\vec{a})+\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})(x_i-a_i)$

$f_{\vec{u}}(\vec{a})=\lim_{h\to0}\frac{f(\vec{a}+h\vec{u})-f(\vec{a})}{h}$

we want to show that

$f(\vec{a}+h\vec{u})\approx L(\vec{a}+h\vec{u})=L(a_1+hu_1,\dots a_n+hu_n)$

$=f(\vec{a})+\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})(\cancel{a_i}+hu_i-\cancel{a_i})$

$=f(\vec{a})+\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})hu_i$

$L(\vec{a}+h\vec{u})-f(\vec{a})=\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})hu_i$

$\lim_{h\to0}\frac{L(\vec{a}+h\vec{u})-f(\vec{a})}{h}=\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})u_i$

$f_{\vec{u}}(\vec{a})=\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\vec{a})u_i=\nabla f(\vec{a})\cdot\vec{u}$

∇ = nabla

where $\vec{u}=(u_1,u_2,\dots,u_n)$

$\nabla f(\vec{a})=\left(\frac{\partial f}{\partial x_1}(\vec{a}),\dots,\frac{\partial f}{\partial x_n}(\vec{a})\right)$ gradient vector in n variables

Directional Derivative via Gradient
If $\mathbf{u}=(u_1,\dots,u_n)$ is a unit vector, the directional derivative of f at $\mathbf{a}$ in the direction of $\mathbf{u}$ is
$$f_{\mathbf{u}}(\mathbf{a})=\nabla f(\mathbf{a})\cdot\mathbf{u}=\sum_{i=1}^n\frac{\partial f}{\partial x_i}(\mathbf{a})u_i.$$

We computed the directional derivative by the inner product with the gradient vector
$$f_{\mathbf{u}}(\mathbf{a})=\nabla f(\mathbf{a})\cdot\mathbf{u}$$
Recall that the inner product measures the alignment of two vectors:
$$\nabla f(\mathbf{a})\cdot\mathbf{u}=\|\nabla f(\mathbf{a})\|\|\mathbf{u}\|\cos\theta=\|\nabla f(\mathbf{a})\|\cos\theta$$
Think, what does this tell us about the gradient vector?

Let $f(x,y,z)=x^2+y^2$. The following picture draws the level surface of $f(x,y,z)=1$. Note how the gradient of f at the point $(0,1,1)$ and $(1,0,1)$ are perpendicular to the level surface.

- Suppose a data set $\{x_1^{(i)},x_2^{(i)},y^{(i)}\}$ with $i=1,\dots,m$ is given.
- The logistic regression model is defined by
$$f(x_1,x_2)=\frac{1}{1+e^{-(\theta_1x_1+\theta_2x_2+b)}}$$
where θ_1 and θ_2 are parameters (weights) for the features. and b is some bias term.
- The predicted outcome using the logistic model is
$$y^{(i)}=f(x_1^{(i)},x_2^{(i)}).$$

Think: What is the range of f?

The amount of error we made is characterized by the following function. Let us define
$$L(\theta_1,\theta_2,b)=-\frac{1}{m}\sum_{i=1}^m\left[y^{(i)}\log(\hat{y}^{(i)})+(1-y^{(i)})\log(1-\hat{y}^{(i)})\right]$$
What is the gradient vector? HW: Check
$$\frac{\partial L}{\partial\theta_1}=\frac{1}{m}\sum_{i=1}^m(\hat{y}^{(i)}-y^{(i)})x_1^{(i)}$$
$$\frac{\partial L}{\partial\theta_2}=\frac{1}{m}\sum_{i=1}^m(\hat{y}^{(i)}-y^{(i)})x_2^{(i)}$$
$$\frac{\partial L}{\partial b}=\frac{1}{m}\sum_{i=1}^m(\hat{y}^{(i)}-y^{(i)})$$

we can use gradient descent

given an initial point x_0, find iterates x_{n+1} recursively using

$x_{n+1}=x_n-\gamma\nabla f(x_n)$

for some $\gamma>0$. where γ (gamma) is the step length

Second order derivatives

example

$f(x,y,z)=x^2y^3z$

$\frac{\partial f}{\partial x}=2xy^3z$

$\frac{\partial f}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)=6xy^2z$

where $\frac{\partial f}{\partial x\partial x}=\frac{\partial f}{\partial x^2}$ = pure second derivative

note that for all mixed partial derivatives

$$\frac{\partial f}{\partial x_i\partial x_j}=\frac{\partial f}{\partial x_j\partial x_i}$$

The hessian Matrix

$$H(f)(x)=\begin{bmatrix}\frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1\partial x_n}\\ \frac{\partial^2 f}{\partial x_2\partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2\partial x_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial^2 f}{\partial x_n\partial x_1} & \frac{\partial^2 f}{\partial x_n\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2}\end{bmatrix}$$

all pure

properties:
- symmetric matrix $H^T=H$ b/c $\frac{\partial f}{\partial x_i\partial x_j}=\frac{\partial f}{\partial x_j\partial x_i}$
- Describes local curvature of f

$$D_v^2(f(x))=D_v(D_v(f(x)))=D_v\left(\sum_{i=1}^n\frac{\partial f}{\partial x_i}v_i\right)=\sum_{i,j=1}^n\frac{\partial f}{\partial x_i\partial x_j}v_iv_j=v^TH(f)(x)v$$

application to optimization

- f has a local max at P_0 if $f(P_0)\ge f(P)$ for all P near P_0
- f has a local min at P_0 if $f(P_0)\le f(P)$ for all P near P_0

these can only occur at critical points where $\nabla f=0$ or undefined or the boundary of f's domain

at a critical point, let $H(f)(P_0)$ be the Hessian Matrix:

use the quadratic form v^THv to test curvature along all directions v:

- $v^THv>0$ for all $v\ne0$ (H is positive definite) → local minimum
- $v^THv<0$ for all $v\ne0$ (H is negative definite) → local maximum
- v^THv can be positive in some direction and negative in others (H is indefinite) → saddle point
- $v^THv=0$ in some $v\ne0$ (H is semidefinite) → test is inconclusive

Positive Definiteness and Eigenvalues

Let $A\in\mathbb{R}^{n\times n}$. A scalar $\lambda\in\mathbb{C}$ is called an eigenvalue of A if there exists a nonzero vector $v\in\mathbb{C}^n$ such that

$$Av=\lambda v$$

this vector (v) is the eigenvector corresponding to λ

note: λ is an eigenvalue if it satisfies:

$$\det(A-\lambda I)=0$$

Theorem (Characterization via Eigenvalues)
Let $H\in\mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\lambda_1,\dots,\lambda_n$. Then:
- H is **positive definite** $\iff$ $\lambda_i>0$ for all i
- H is **negative definite** $\iff$ $\lambda_i<0$ for all i
- H is **indefinite** $\iff$ it has both positive and negative eigenvalues
- H is **semidefinite** $\iff$ all eigenvalues are nonnegative (positive semidefinite) or nonpositive (negative semidefinite)

At a critical point P_0, let $H=H(f)(P_0)$ be the Hessian matrix: Compute eigenvalues of H.
- All positive eigenvalues $\implies$ local minimum
- All negative eigenvalues $\implies$ local maximum
- Mixed signs $\implies$ saddle point
- Zero eigenvalue(s) $\implies$ test inconclusive

example:

find and ~~analyse~~ any critical points of $f(x,y)=-\sqrt{x^2+y^2}$

$=-(x^2+y^2)^{\frac12}$

$\nabla f(x,y)=\left(\frac{\partial f}{\partial x}(x^2+y^2)^{\frac12},\frac{\partial f}{\partial y}(x^2+y^2)^{\frac12}\right)$

$=\left(-\frac12(x^2+y^2)^{-\frac12}2x,-\frac12(x^2+y^2)^{-\frac12}2y\right)$

$=\left(-\frac{x}{\sqrt{x^2+y^2}},-\frac{y}{\sqrt{x^2+y^2}}\right)$

we can tell that $(0,0)$ is the critical point (because at $(0,0)$, ∇f is undefined)

So...

$$H(f)(x)=\begin{bmatrix}\frac{-\sqrt{x^2+y^2}+\frac12(x^2+y^2)^{-\frac12}2x}{x^2+y^2} & -x(-\frac12)(x^2+y^2)^{-\frac12}\\ -x(-\frac12)(x^2+y^2)^{-\frac12} & \frac{-\sqrt{x^2+y^2}+\frac12(x^2+y^2)^{-\frac12}2x}{x^2+y^2}\end{bmatrix}$$

$$H(f)(0,0)=\begin{bmatrix}-\frac{1}{\sqrt{x^2+y^2}} & 0\\ 0 & -\frac{1}{\sqrt{x^2+y^2}}\end{bmatrix}$$

Since $-\frac{1}{\sqrt{x^2+y^2}}$ is undefined for $(x,y)=(0,0)$

we can't use the second derivative test

example: x^2-y^2

$\nabla f=(2x,2y)=0$, when $x=y=0$

$H(f)(x)=\begin{bmatrix}2&0\\0&-2\end{bmatrix}$

we know that

$\det\left(\begin{bmatrix}2-\lambda&0\\0&-2-\lambda\end{bmatrix}\right)=0$

$\iff(2-\lambda)\cdot(2+\lambda)=0\iff\lambda=2$ or $\lambda=-2$

So saddle point

Theorem
If f is continuous on a closed and bounded region R, then f attains both a global maximum and minimum somewhere in R.
- Closed region: contains its boundary
- Bounded region: does not stretch to infinity

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chain rule:

$z=f(x,y)$

where: $x=g(t)$, $y=h(t)$

So... $z=f(g(t),h(t))$

$$\frac{dz}{dt}=\frac{\partial f}{\partial x}\cdot\frac{dx}{dt}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dt}$$

example:

$z=f(x,y)=x\sin y$

$x=t^2$

$y=2t+1$

$z=t^2\sin(2t+1)$

$\frac{dz}{dt}=2t\cdot\sin(2t+1)+t^2\cos(2t+1)\cdot2$

$\frac{\partial f}{\partial x}=\sin y$

$\frac{\partial f}{\partial y}=x\cos y$

$\frac{dx}{dt}=2t$

$\frac{dy}{dt}=2$

So... $\sin(2t+1)2t+x\cos(2t+1)2$

for $z=f(x,y)$, $x=g(u,v)$, $y=h(u,v)$

$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$

$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$

or for n variables

$\frac{\partial z}{\partial t_j}=\sum_{i=1}^n\frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t_j}$, $j=1,\dots,m$

if f is differentiable, then all partial derivatives exist there (f has linearization)

$L(x)=f(a)+\sum_{i=1}^n m_i(x_i-a_i)$

we want to show that $m_i=\frac{\partial f}{\partial x_i}(a)$

$\lim_{h\to0}\frac{E(a+h)}{\|h\|}=0$

$E(x)=f(x)-L(x)$

$E(a+h)=f(a+h)-L(a+h)$

$=f(a+h)-f(a)+\sum_{i=1}^n m_i(a_i+h_i-a_i)$

because of $\lim_{h\to0}\frac{f(a_1\dots a_i+h_i\dots a_n)-(a_1\dots a_n)}{\|h\|}-m_i=0$

note: even if all derivatives exists at a point, this does not guarantee differentiability

consider this example:

$f(x,y)=x^{\frac13}y^{\frac13}$

$\frac{\partial f}{\partial x}=\frac13x^{-\frac23}y^{\frac13}$

$\frac{\partial f}{\partial y}=\frac13x^{\frac13}y^{-\frac23}$

$f(x,y)=\ln(x^2+y^2)$

domain: $\mathbb{R}^2\setminus\{(0,0)\}$

$\frac{\partial f}{\partial x}=\frac{2x}{x^2+y^2}$ cont everywhere $(0,0)$

$\frac{\partial f}{\partial y}=\frac{2y}{x^2+y^2}$ cont everywhere $(0,0)$

so f is differentiable on its domain

Intergration

f(x)
a
b
$\Delta x=\frac{b-a}{n}$

$\sum_{k=1}^n$ Area of rectangle $k=f(x_k^*)\cdot\Delta x$

we take $n\to\infty$

$\lim_{n\to\infty}\sum_{k=1}^n f(x_k)\Delta x=\int_a^b f(x)\,dx$

if this limit exists (no point is undefined in the interval) we say f is integrable on $[a,b]$

evaluate $\int_0^3 2x\,dx$ as a limit of Riemann sums

$=\lim_{n\to\infty}\sum_{k=1}^n 2x_k\frac{3}{n}$

$=\lim_{n\to\infty}\frac{6}{n}\sum_{k=1}^n x_k$

Theorem
If f is continuous on $[a,b]$, then
$$\int_a^b f(x)\,dx=F(b)-F(a),$$
where F is any antiderivative of f, i.e., $F'(x)=f(x)$.

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for $f(x,y)=x^{\frac13}y^{\frac13}$

$f_x(0,0)=\lim_{h\to0}\frac{f(0+h,0)-f(0,0)}{h}=\lim_{h\to0}\frac{0}{h}=\lim_{h\to0}\frac{0}{1}=0$ use L'Hôpital's rule

Similarly $f_y(0,0)=0$

Both derivatives exists at $(0,0)$

Suppose f is differentiable then the linear approx at $(0,0)$ would have to be $L(x,y)=0$

but $\lim_{(h_1,h_2)\to0}\frac{f(h_1,h_2)-L(h_1,h_2)}{\sqrt{h_1^2+h_2^2}}=\lim_{h_1,h_2\to0}\frac{h_1^{\frac13}h_2^{\frac13}}{\sqrt{h_1^2+h_2^2}}=\lim_{h\to0}\frac{h^{\frac23}}{\sqrt{2h^2}}=\lim_{h\to0}\frac{h^{\frac23}}{\sqrt2 h}=\lim_{h\to0}\frac{1}{\sqrt2}\frac{1}{h^{\frac13}}=DNE$

this is contradiction → f is not differentiable

$F=\int f(x)\,dx$ is the antiderivative of f

if F is an antiderivative of f then $F+c$, where c is a constant is also an antiderivative

$F(b)-F(a)=(F(x_n)-F(x_{n-1}))+\dots+(F(x_1)-F(x_0))$

$=\sum_{k=1}^n F(x_k)-F(x_{k-1})$

$=\sum_{k=1}^n F'(x_k)\cdot\Delta x$ (mean value theorem)

$=\sum_{k=1}^n f(x_k)\cdot\Delta x$

taking n to infinity $\lim_{n\to\infty}\sum_{k=1}^n f(x_k)\cdot\Delta x=\int_a^b f(x)\,dx$

thus $F(b)-F(a)=\int_a^b f(x)\,dx$

$\frac{d}{dx}\int_a^x\cos t\,dt=\cos x$

note that: $\int_a^b f(x)dx=-\int_b^a f(x)dx$

example

$g(x)=\int_1^x\sqrt{1+t^3}\,dt$

then $\frac{d}{dx}g(x)=\sqrt{1+x^3}$

and $g(x^3)=\int_1^{x^3}\sqrt{1+t^3}\,dt$ ← chain rule

$\frac{d}{dx}g(x^3)=\frac{d}{dx}g(x^3)3x^2$

$=\sqrt{1+x^9}\cdot3x^2$

Proof of Second FTC
Suppose $g(x)=\int_a^x f(t)dt$. Then
$$\frac{g(x+h)-g(x)}{h}=\frac{\int_x^{x+h}f(t)dt}{h}.$$
Since f is continuous on $[x,x+h]$, by the Extreme Value Theorem, f attains minimum m and maximum M on $[x,x+h]$, so
$$mh\le\int_x^{x+h}f(t)dt\le Mh.$$
Dividing by h gives
$$m\le\frac{\int_x^{x+h}f(t)dt}{h}\le M.$$
As $h\to0$, $m\to f(x)$ and $M\to f(x)$, so by the Squeeze Theorem,
$$\frac{d}{dx}\left[\int_a^x f(t)dt\right]=f(x).$$

$\int f(x)\,dx$: antiderivative

$\int_a^b f(x)\,dx$: definite integral (real number)

$\int_a^x f(x)\,dx$: is a function of x

$\int_a^\infty f(x)\,dx$: $\lim_{b\to\infty}\int_a^b f(x)dx$ if limit exists

example compute $\int_0^\infty e^{-5x}\,dx$

$=\lim_{b\to\infty}\int_0^b e^{-5x}dx$

chain rule

$=\lim_{b\to\infty}-\frac15e^{-5x}\Big|_0^b$

$=\lim_{b\to\infty}\left(-\frac15e^{-5b}-(-\frac15e^0)\right)$

$\lim_{b\to\infty}\left(\cancel{-\frac15e^{-5b}}+\frac15\right)$

$=\frac15$

u-substitution

chainrule...

$\frac{d}{dx}f(g(x))=f'(g)\cdot g'$

this tells us that

$\int f'(g)g'\,dx=f(g)+c$

example

$\int_2^3 2x\sin x^2\,dx$

let $u=x^2$ then $du=2x\,dx$

$\int_{u=4}^{u=9}\sin u\,du=-\cos u\Big|_4^9$

$=-\cos9-(-\cos4)$

$=-\cos9+\cos4$

examples

eval

① $\int_0^1 x(x^2+1)^4\,dx$

$u=x^2+1$

$du=2x\,dx$ $\frac12du=x$

$=\int_{u=1}^{u=2}\frac12u^4du$

$=\frac12\int_1^2u^4du$

$=\frac12\left(\frac15u^5\Big|_1^2\right)=\frac12\cdot\frac15(2^5-1)=\frac{1}{10}(31)=3.1$

② $\int\sin^2x\,dx$ $\sin^2x=\frac{1-\cos(2x)}{2}$

$=\int\frac{1-\cos(2x)}{2}dx$

$=\int\frac12dx-\int\frac{\cos(2x)}{2}dx$ now $\int\frac{\cos(2x)}{2}dx$

$=\frac12x-$ let $u=2x$, $du=2dx$, $\frac12du=\frac22dx=\frac12dx$

Integration by part

Recall the productrule

$\frac{d}{dx}fg=f'g+fg'$

now integrate

$\int f'g\,dx+\int fg'\,dx=fg+c$

and so..

$\int fg'\,dx=fg-\int f'g\,dx$

example:

① $\int_0^1 xe^x\,dx$

say that...

$u=e^x dx$ $dv=x$

since

$\int u\cdot dv=uv-\int v\cdot du$

$du=e^x$ and $v=\frac12x^2$

then

$\int_0^1 xe^x\,dx=uv\Big|_0^1-\int_0^1 v\,du=xe^x\Big|_0^1-ex\Big|_0^1$

$=e-(e-1)$

$=e-e+1=1$

② $\int_1^2\ln(x)\,dx=\int_1^2 1\cdot\ln(x)\,dx$

let $u=1$ and $dv=\ln(x)dx$ ✗

$du=dx$ and $v=\int\ln(x)$ ← we can't do

try again ^^

$u=\ln(x)$ and $dv=1dx$

extending to two variables

let $\Delta=\frac{b-a}{n}$

then

$\int_a^b f(x)\,dx=\lim_{n\to\infty}\sum_{i=1}^n f(x_i)\cdot\Delta$

let $f(x,y)$ be defined on $a\le x\le b$, $c\le y\le d$

devide both intervals into n & m

$\Delta x=\frac{b-a}{n}$, $\Delta y=\frac{d-c}{m}$

$\sum_{j=1}^m\sum_{i=1}^n f(x_{ij})\cdot\Delta x\Delta y\approx$ volume under $f(x)$

note that this ↓ must remain true

$$\sum_{i=1}^n\sum_{j=1}^m L_{ij}\Delta x\Delta y\le\sum_{i=1}^n\sum_{j=1}^m f(u_{ij},v_{ij})\Delta x\Delta y\le\sum_{i=1}^n\sum_{j=1}^m M_{ij}\Delta x\Delta y$$

then as $n,m\to\infty$..

$$\lim_{n,m\to\infty}\sum_{i=1}^n\sum_{j=1}^m f(u_{ij},v_{ij})\,\Delta x\Delta y=\int_R f\,dA$$

$[a,b]\times[c,d]$; $dxdy$ because $\Delta x\cdot\Delta y=A$

alternate notation

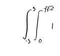
$$\int_a^b\int_c^d f\,dx\,dy$$

① $g(x,y)=1$, then $\int_R g\,dA$ = just the area

②

A city occupies a semicircular region of radius 3 km bordering the ocean

find the average distance of points in the city to the ocean



$x^2+y^2\le3^2$

$y=\sqrt{3^2-x^2}$

$$\int_{-3}^3\int_0^{\sqrt{3^2-x^2}}y\,dy\,dx=\int_{-3}^3\frac12y^2\Big|_0^{\sqrt{3^2-x^2}}dx=\int_{-3}^3\frac12(3^2-x^2)==\frac12\left(9x-\frac13x^3\right)\Big|_{-3}^3$$

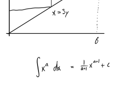
$=\frac12(27-9+27-9)$

$=18$

$$\int_{-3}^3\int_0^{\sqrt{3^2-x^2}}1\,dy\,dx=\frac12\pi3^2=\frac92\pi$$

in total the avg dist is $\frac{18}{\frac92\pi}=\frac{4}{\pi}$ km

example



$\int x^a\,dx=\frac{1}{a+1}x^{a+1}+c$

Show that you can swap integration

$$\int_0^6\int_{x/3}^2 x\sqrt{y^3+1}\,dy\,dx$$

much easier ↙

$$\int_0^2\int_0^{3y}x\sqrt{y^3+1}\,dx\,dy$$

$=\int_0^2\frac12x^2\sqrt{y^3+1}\Big|_0^{3y}dy$

$=\int_0^2\frac12 9y^2\sqrt{y^3+1}\,dy$

$u=y^3+1$

$du=3y^2\,dy$

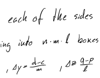
$\int_1^9\frac12 3\sqrt{u}\,du$

$\frac32\int_1^9\sqrt{u}\,du$

$=u^{\frac32}\Big|_1^9$

$=9^{\frac32}-1=27-1=26$

We can define integrals over a solid region W in $\mathbb{R}^3$ by subdividing into boxes.



Using similar reasoning, express $\int_W f\,dV$ as a limit of Riemann sums.

Let us divide each of the sides into n,m and ℓ pieces

we are chopping into $n\cdot m\cdot\ell$ boxes

Let $\Delta x=\frac{b-a}{n}$, $\Delta y=\frac{d-c}{m}$, $\Delta z=\frac{q-p}{\ell}$

$\Delta V=\Delta x\cdot\Delta y\cdot\Delta z$ is the volume of each smaller box

Pick a point $(u_{ijk},v_{ijk},w_{ijk})$ in the ijk-th small box

$$\lim_{n,m,\ell\to\infty}\sum_{ijk}f(u_{ijk},v_{ijk},w_{ijk})\cdot\Delta x\Delta y\Delta z=\int_W f\,dV=\int_p^q\int_c^d\int_a^b f(x,y,z)\,dx\,dy\,dz$$

$\sum_{ijk}=\sum_{i=1}^n\sum_{j=1}^m\sum_{k=1}^\ell$

note that $\Updownarrow$

$$\int_a^b\int_c^d\int_p^q f(x,y,z)\,dz\,dy\,dx$$

f is integratable if the limit exists

Compute $\int_W(1+xyz)\,dV$ over the cube $0\le x,y,z\le4$

$=\int_0^4\int_0^4\int_0^4(1+xyz)\,dx\,dy\,dz$

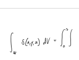
$\int_0^4\int_0^4 x+\frac12x^2yz\Big|_0^4\,dy\,dz$

$\int_0^4\int_0^4 4+8yz\,dy\,dz$

$\int_0^4 4y+4y^2z\Big|_0^4\,dz$

$\int_0^4 16+64z\,dz=16z+32z^2\Big|_0^4=64+512=576$

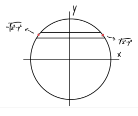
Set up an iterated integral for the mass of a solid cone bounded by $z=\sqrt{x^2+y^2}$ and $z=3$, with density $\delta(x,y,z)=z$.



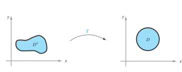
$\int_W\delta(x,y,z)\,dV=\int_0^3\int\int$

$z=\sqrt{x^2+y^2}\iff z^2=x^2+y^2$

$$\int_{-z}^{z}\int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}}\delta(x,y,z)\,dx\,dy\,dz$$



- Think of how T deforms subsets of D^* rather than the graph.
- **Image set:** $T(D^*)=\{(x,y):(x,y)=T(u,v)$ for some $(u,v)\in D^*\}$.



example

Let $D^*=[0,1]\times[0,2\pi]$ with coordinates (r,θ) and

$T(r,\theta)=(r\cos\theta,r\sin\theta)$

what is $T(D^*)$

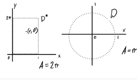
let $x=r\cos\theta$

$y=r\sin\theta$

Since $x^2+y^2=r^2(\cos^2\theta+\sin^2\theta)=r^2$

Since $0\le r\le1$, then $(x,y)\in D$ has the property

s.t. $x^2+y^2\le1$, which is a unit disk



Theorem (linear map)
If A is 2×2 matrix with $\det A\ne0$ and $T(u)=Au$, then T maps parallelograms to parallelograms and is bijective.

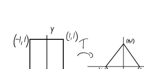
Example: $T(u,v)=(\frac{u+v}{2},\frac{u-v}{2})$ on $D^*=[-1,1]^2$. Find $T(D^*)$.

Let $T(u,v)=\left(\frac{u+v}{2},\frac{u-v}{2}\right)$ on $D^*=[-1,1]^2$

$$\underbrace{\begin{pmatrix}\frac12&\frac12\\\frac12&-\frac12\end{pmatrix}}_{A}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}\frac{u+v}{2}\\\frac{u-v}{2}\end{pmatrix}$$

$\det(A)=-\frac14-\frac14=-\frac12$

$|\det(A)|=\frac12$



For $T(u,v)=(x(u,v),y(u,v))$ define

$$J(u,v)=\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}x_u&x_v\\y_u&y_v\end{vmatrix}.$$

Provided the partial derivatives exist.

Consider $T(u,v)=A\begin{pmatrix}u\\v\end{pmatrix}$ where

$$A=\begin{pmatrix}a&b\\c&d\end{pmatrix}.$$

Then

$$x=au+bv,\qquad y=cu+dv.$$

Compute the Jacobian matrix:

$$J=\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}\frac{\partial x}{\partial u}&\frac{\partial x}{\partial v}\\\frac{\partial y}{\partial u}&\frac{\partial y}{\partial v}\end{vmatrix}=\begin{vmatrix}a&b\\c&d\end{vmatrix}=A.$$

Hence for a linear map $T=Au$, the Jacobian matrix is A.

---

use pl co din es o compute

$$\left(\int_{-\infty}^{\infty}e^{-x^2}dx\right)^2=\left(\int_{-\infty}^{\infty}e^{-x^2}dx\right)\left(\int e^{-x^2}dx\right)$$

$$=\left(\int_{-\infty}^{\infty}e^{-y^2}dy\right)\left(\int_{-\infty}^{\infty}e^{-y^2}dy\right)$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{-x^2}e^{-y^2}\,dx\,dy$$

$$=\int_{\mathbb{R}^2}e^{-(x^2+y^2)}\,dx\,dy$$

$$=\lim_{a\to\infty}\int_0^{2\pi}\int_0^a e^{-r^2}r\,dr\,d\theta=\lim_{a\to\infty}\int_0^{2\pi}-\frac12e^{-r^2}\Big|_0^a\,d\theta=\lim_{a\to\infty}\int_0^{2\pi}\left(\frac12e^{-a^2}+\frac12\right)d\theta=\lim_{a\to\infty}\left(\cancel{-\pi e^{-a^2}}+\pi\right)=\pi$$

Let $x=x(r,\theta,z)=r\cos\theta$

$y=y(r,\theta,z)=r\sin\theta$

$z=z(r,\theta,z)=z$